

Cross-linking/MS-analysis of Thr72-phosphorylated Rab8A and PPM1H(D288A) complex

Authors: Pawel Lis, Pui Yiu Lam, Axel Knebel, Kerryn Berndsen, Dario R Alessi

Medical Research Council Protein Phosphorylation and Ubiquitylation Unit, School of Life Sciences, University of Dundee, Dow Street, Dundee DD1 5EH, UK

Correspondence to Pawel Lis (p.lis@dundee.ac.uk) or Dario R Alessi (d.r.alessi@dundee.ac.uk)

Summary:

A subset of Rab proteins, including Rab8A, have been identified as substrates of the Leucine Rich Repeat Kinase 2 (LRRK2) and the Protein Phosphatase PPM1H (Steger et al. 2017; Berndsen et al. 2019). As the crystal structure of the complex of PPM1H with Rab8A pT72 is unavailable, we employed a cross-linking approach (Iacobucci et al. 2018) to perform a study to identify the topology of this phosphatase-substrate binding. A 'substrate-trapping' mutant (D288A) of PPM1H was identified (Berndsen et al. 2019), and used in this study to form stable complex of PPM1H with Rab8A pT72. We then used Disuccinimidyl Dibutyric Urea (DSBU) to cross-link the complex of PPM1H (D288A) 'substrate-trapping' mutant and phospho-Rab8A. DSBU is a mass spectrometry cleavable amine reactive crosslinker with a 12.5 Å spacer arm that is widely used to identify and map sites of protein-protein interactions (Pan et al. 2018). This reagent cross-links Lys residues to acidic and hydroxyl amino acids located within 32 Å (Götze et al. 2019). Crosslinked samples were excised from a Coomassie-stained poly acrylamide gel and digested with 3 conditions (trypsin, trypsin/Asp-N and trypsin/Glu-C). In addition, SCX cartridge purification was applied in one of the tryptic digested samples to further enrich the cross-linked peptides. PPM1H and pRab8a cross-linked peptides were identified using meroX software (Götze et al. 2012). Potential crosslinked peptides with score higher than 50 as well as false discovery rate (FDR) less than 5% were manually inspected to confirm only a single crosslinked site was proposed from each peptide (Iacobucci et al. 2018).

Materials

1.1. Consumables:

1.1.1. Recombinant PPM1H [D288A] phosphatase (Purified as described in protocols.io [dx.doi.org/10.17504/protocols.io.bu7wnzpe](https://doi.org/10.17504/protocols.io.bu7wnzpe) or purchased <https://mrccpureagents.dundee.ac.uk/> (DU68087) in 40 mM HEPES pH 7.5 or PBS. Note it is critical that no Tris or other amine containing buffers are present in the protein prep, as this could interfere with the assay. A buffer exchange into 40 mM HEPES pH 7.5, 150 mM NaCl, 2 mM MgCl₂ buffer is needed prior to the

cross-linking (described as the first step of this protocol). Wild-type PPM1H phosphatase (DU62835) can be used as a control.

1.1.2. Recombinant stoichiometrically Thr72 phosphorylated Rab8A (described in protocols.io [dx.doi.org/10.17504/protocols.io.butinwke](https://doi.org/10.17504/protocols.io.butinwke) or purchased <https://mrcppureagents.dundee.ac.uk/> (DU68198). Note it is critical that no Tris or other amine containing buffers are present in the protein prep, as this could interfere with the assay. A buffer exchange into 40 mM HEPES pH 7.5, 150 mM NaCl, 2 mM MgCl₂ buffer is needed prior to the cross-linking (described as the first step of this protocol). Non-phosphorylated Rab8A can be used as a control.

1.1.3. Tris Base (VWR Chemicals, 103157P)

1.1.4. Hepes (VWR Chemicals, 441487M)

1.1.5. Magnesium Chloride hexahydrate (Sigma M2670)

1.1.6. Sodium chloride (VWR 27810.364)

1.1.7. DSBU (Disuccinimidyl Dibutyric Urea) (Thermo Scientific™, A35459)

1.1.8. Dimethyl sulfoxide anhydrous, >= 99.9 % | 67-68-5 | Sigma-Aldrich?

1.1.9. Sequencing grade trypsin (Promega, V5111)

1.1.10. Asp-N (Promega, V1621)

1.1.11. Endoproteinase Glu-C Sequencing Grade (Merck, 11420399001)

1.1.12. ThermoFisher NP0321BOX 4-12% NuPAGE Bis-Tris 1mm 10-well gel

1.1.13. Colloidal Blue Staining Kit (Invitrogen™ LC6025)

1.1.14. NuPAGE™ LDS Sample Buffer (4X) (Invitrogen, NP0008)

1.1.15. Iodoacetamide (IAA, Sigma, I1149)

1.1.16. Acetonitrile (ACN, LC grade; MERCK; cat. no. 1.00030.2500)

1.1.17. Ammonium Acetate (NH₄OAc; BioUltra grade; Sigma-Aldrich; cat. no. 09688)

1.1.18. Formic acid (FA; MERCK; cat. no. 1.00263.1000)

1.1.19. Methanol (MeOH; LC-MS grade; MERCK; cat. no. 1.06035.1000)

1.1.20. Acetic acid glacial (VWR, 36289.K3)

1.1.21. Phosphoric acid (H₃PO₄; 85% (v/v) solution; Sigma-Aldrich; cat. no. 345245)

1.1.22. Trifluoroacetic acid (TFA; HPLC grade; MERCK; cat. no. 302031)

1.1.23. Ammonium bicarbonate (ABC) (Sigma, 09830-500G)

1.2. Solutions (all made fresh)

1.2.1. 40 mM HEPES pH 7.5, 150 mM NaCl, 2 mM MgCl₂ in milli-Q water

1.2.2. 1 M Tris pH 8.8 in milli-Q water

1.2.3. 100 mM ABC (Ammonium bicarbonate) in milli-Q water

1.2.4. 25 mM ABC (Ammonium bicarbonate) in milli-Q water

1.2.5. 10 mM DTT in milli-Q water

1.2.6. 55 mM IAA in 100 mM ABC

1.2.7. 50 mM ABC in 50% (v/v) acetonitrile (prepared by mixing equal volumes of 100 mM ABC and ACN)

1.2.8. 5% (v/v) TFA in milli-Q water

- 1.2.9. 1.67% (v/v) TFA in acetonitrile (prepared by mixing 2 volumes of ACN with 1 volume of 5% (v/v) TFA solution)

1.3. Equipment

- 1.3.1. 2 µl, 20 µl, 200 µl and 1000 µl Gilson pipettes
- 1.3.2. 1.5 ml protein low bind Eppendorf tubes (Eppendorf™ #022431081)
- 1.3.3. Diacon 0.5 ml 10 kDa MWCO dialysis tube (MD6-71, Molecular Dimensions)
- 1.3.4. Eppendorf Thermomixer
- 1.3.5. Magnetic stirrer with stir bars
- 1.3.6. Nanodrop 1000 (Thermo Fisher Scientific)
- 1.3.7. SpeedVac (SPD140DDA; Thermo Fisher Scientific)
- 1.3.8. MCX Cartridge (OASIS PRiME MCX 1 cc cartridge; Waters Corporation)
- 1.3.9. HPLC analytical column (PepMap™ RSLC C18, 75 µm x 50 cm, 2 µm, 100 Å; Thermo Fisher Scientific)
- 1.3.10. HPLC trap column (Acclaim PepMap™ 100, C18, 100 µm x 2 cm, 5 µm, 100 Å; Thermo Fisher Scientific)
- 1.3.11. HPLC system (Ultimate 3000 RSLC nano-HPLC; Thermo Fisher Scientific)
- 1.3.12. Mass spectrometer (Orbitrap Exploris™ 480; Thermo Fisher Scientific)
- 1.3.13. Water bath (Bandelin Electronic)

1.4. Software

- 1.4.1. MeroX software 2.0

2. Method

2.1. Protein buffer exchange

As the cross-linking reaction can be inhibited by amine containing buffers, a buffer exchange step for both, PPM1H and pRab8A protein solutions is needed prior to the cross-linking. Proteins should be dialysed into 40 mM HEPES pH 7.5, 150 mM NaCl, 2 mM MgCl₂ buffer.

- 2.1.1. Prepare a beaker containing 500 ml of 40 mM HEPES pH 7.5, 150 mM NaCl, 2 mM MgCl₂ buffer and a magnetic stir bar. Set the beaker on a magnetic stirrer with gentle stirring, at RT.
- 2.1.2. Pipette up to 0.5 ml of each protein into separate Diacon dialysis tubes. Put the tubes in a floating rack, and place the rack on the surface of the buffer in the beaker. Incubate for 2 hours at RT.
- 2.1.3. Transfer the floating rack with the dialysis tubes into a fresh beaker containing 500 ml of 40 mM HEPES pH 7.5, 150 mM NaCl, 2 mM MgCl₂ buffer and a magnetic stir bar. Set the beaker on a magnetic stirrer with gentle stirring, at RT. Incubate overnight at RT.

2.2. Preparation of the cross-linked pRab8A/PPM1H complex samples

- 2.2.1. Prepare a solution of stoichiometrically Thr72 phosphorylated Rab8A at a concentration 35 μ M (~0.7 mg/ml, migrates at 20 kDa) and PPM1H at 28 μ M (~1.4 mg/ml, migrates at 50 kDa) in total volume of 15.5 μ l using 40 mM HEPES pH 7.5, 150 mM NaCl, 2 mM $MgCl_2$ as a dilution buffer, to achieve 1.25-fold molar excess of Rab8A to PPM1H. The mixed volumes will depend on protein concentrations obtained after the buffer exchange step.
- 2.2.2. Mix the protein solution gently and incubate at 30°C for 1 h.
- 2.2.3. After this incubation incubate at room temperature for 10 min.
- 2.2.4. During this time prepare a fresh 300 mM stock solution of the cross-linker by dissolving 1 mg of DSBU in 15.5 μ l of anhydrous DMSO, vortex briefly, keep on ice, use within 5 minutes.
- 2.2.5. Add 0.5 μ l of the cross-linker solution into 15.5 μ l of the protein solution, mix gently. This will result in final concentration of 9.375 mM of DSBU (250-fold molar excess to pRab8A, 335-fold molar excess to PPM1H) in 16 μ l of reaction mix.
- 2.2.6. Incubate the cross-linking reaction for 10 min at room temperature. Add 2 μ l of 1 M Tris pH 8.8 to quench the reaction, mix gently.
- 2.2.7. Add 7 μ l of 4 x NuPAGE™ LDS Sample Buffer (containing lithium dodecyl sulfate at pH 8.5 with SERVA Blue G250 and phenol red) (do not add reducing agent such as 2-mercaptoethanol or heat the samples before electrophoresis as this could impact crosslinking reactions), vortex. Immediately resolve the samples using SDS-PAGE.

2.3. SDS-PAGE and in-gel digestion

- 2.3.1. Resolve the samples on a 4-12% gradient NuPAGE Bis-Tris gel. The final reaction volume is 25 μ l. Load 6 μ l aliquots of each sample into 4 lanes (one for each protease condition: Sample A – trypsin, Sample B – trypsin+SCX, Sample C – trypsin+AspN, Sample D – trypsin+GluC). Run the gel at 70 V for 20 minutes, and then at 120 V until the dye front leaves the gel.
- 2.3.2. Fix and stain the gel using Invitrogen™ Colloidal Blue Staining Kit
 - 2.3.2.1. Prepare the fixing solution (10% (v/v) glacial acetic acid in 50% (v/v) methanol) by mixing 5 ml of acetic acid with 25 ml of methanol and 20 ml of milli-Q water.
 - 2.3.2.2. Incubate the gel in the fixing solution for 10 minutes at RT with gentle shaking. Discard the fixing solution.
 - 2.3.2.3. Prepare the staining solution (20% (v/v) methanol, 20% (v/v) Stainer A, 5% (v/v) Stainer B). Mix 10 ml of Stainer A with 10 ml of methanol and 22.5 ml of milli-Q water. Do not add Stainer B at this stage. Incubate the gel in the solution for 10 minutes at RT with gentle shaking.

- 2.3.2.4. Add 2.5 ml of Stainer B directly to the solution containing the gel. Incubate the gel in the staining solution for 2 – 4 h at RT with gentle shaking.
- 2.3.2.5. Rinse the gel using milli-Q water at least 3 times within at least 4 hours or overnight.
- 2.3.2.6. Image the gel.

Note: Band-shifts corresponding to a PPM1H dimer size (100 kDa), and a tetramer of PPM1H and pRab8A (140 kDa) are an indication of a successful cross-linking (Figure 1). Decide which crosslinked species need to be excised. We excised the pRab8A-PPM1H crosslinked species and the PPM1H dimer crosslinked species highlighted on Figure 1.

- 2.3.3. Prepare low-binding 1.5 ml Eppendorf tubes – one for each condition (A, B, C, D).
- 2.3.4. Using a fresh, sterile scalpel, excise each band of interest, cut into 1mm cubes, and place the gel pieces in a.
- 2.3.5. Add 500 µl of acetonitrile and incubate at room temperature for 10 minutes to dehydrate the gel pieces.
- 2.3.6. Discard the supernatant.
- 2.3.7. Add 50 µl of freshly prepared 10 mM DTT solution and ensure gel pieces are covered. If necessary, add an additional 50 µl of 10 mM DTT to cover gel pieces.
- 2.3.8. Incubate in a Thermomixer at 56°C for 30 min with mixing (1200 rpm).
- 2.3.9. Add 500 µl of acetonitrile.
- 2.3.10. Incubate at room temperature for 10 minutes without shaking to dehydrate the gel pieces.
- 2.3.11. Discard the supernatant.
- 2.3.12. Prepare a fresh solution of 55 mM Iodoacetamide (IAA) in 100 mM ABC. Please note that iodoacetamide is toxic and should be carefully weighed out in a fume hood (for details refer to the safety data sheet).
- 2.3.13. Add 50-100 µl of freshly prepared 55 mM iodoacetic acid solution to ensure that gel pieces are covered.
- 2.3.14. Cover reaction mixture from light by covering the tubes with aluminium foil.
- 2.3.15. Incubate in a Thermomixer at room temperature for 20 min with mixing (1200 rpm).
- 2.3.16. After this incubation add 500 µl of acetonitrile.
- 2.3.17. Incubate at room temperature for 10 minutes to dehydrate the gel pieces without shaking.
- 2.3.18. Discard the supernatant. Please note that the solution contains toxic iodoacetamide and should be disposed of in an appropriate manner. For details refer to the safety data sheet.
- 2.3.19. Add 100 µl of 50 mM ABC in 50% (v/v) acetonitrile
- 2.3.20. Incubate at room temperature for 30 min without shaking.

- 2.3.21. Add 500 μ l of acetonitrile. Do not discard the supernatant from the previous step.
- 2.3.22. Incubate at room temperature for 10 minutes to dehydrate the gel pieces.
- 2.3.23. Discard the supernatant.
- 2.3.24. Prepare fresh 80 μ l solution of 50 ng/ μ l Trypsin in milli-Q water keep on ice.
- 2.3.25. Prepare fresh 20 μ l solution 20 ng/ μ l, AspN in milli-Q water keep on ice.
- 2.3.26. Prepare fresh 20 μ l solution 50 ng/ μ l, GluC in milli-Q water keep on ice.
- 2.3.27. 4 different protease digestion conditions were used: Sample A – trypsin, Sample B – trypsin+SCX enrichment, Sample C – trypsin+AspN, Sample D – trypsin+GluC.
- 2.3.28. For A add 20 μ l of Trypsin solution to the dehydrated gel pieces. If the gel pieces are not covered completely, add 25 mM ABC solution to cover them.
- 2.3.29. For B add 20 μ l of Trypsin solution to the dehydrated gel pieces. If the gel pieces are not covered completely, add 25 mM ABC solution to cover them.
- 2.3.30. For C add 20 μ l of Trypsin solution to the dehydrated gel pieces, followed by 20 μ l of AspN solution. If the gel pieces are not covered completely, add 25 mM ABC solution to cover them.
- 2.3.31. For D add 20 μ l of Trypsin solution to the dehydrated gel pieces, followed by 20 μ l of GluC solution. If the gel pieces are not covered completely, add 25 mM ABC solution to cover them.
- 2.3.32. Incubate all 4 samples overnight at 30°C with mixing (1200 rpm).
- 2.3.33. Prepare fresh peptide elution solution (1.67% (v/v) TFA in acetonitrile) by mixing 2 volumes of ACN with 1 volume of 5% (v/v) TFA solution.
- 2.3.34. Add 100 μ l peptide elution solution to each of the digestion tubes.
- 2.3.35. Incubate at 37°C for 15 min with mixing (1200 rpm).
- 2.3.36. Transfer the supernatant that contains the digested protein into a fresh 1.5 ml low-binding Eppendorf tube. Avoid collecting any gel pieces.
- 2.3.37. In order to extract the remaining peptides present in the gel pieces, repeat the elution step by adding another 100 μ l into the gel pieces
- 2.3.38. incubate at 37°C for 15 min.
- 2.3.39. Collect the supernatant and combine with the previously collected supernatant in step 2.3.36.
- 2.3.40. Freeze the samples on dry ice,
- 2.3.41. Vacuum dry using SpeedVac.

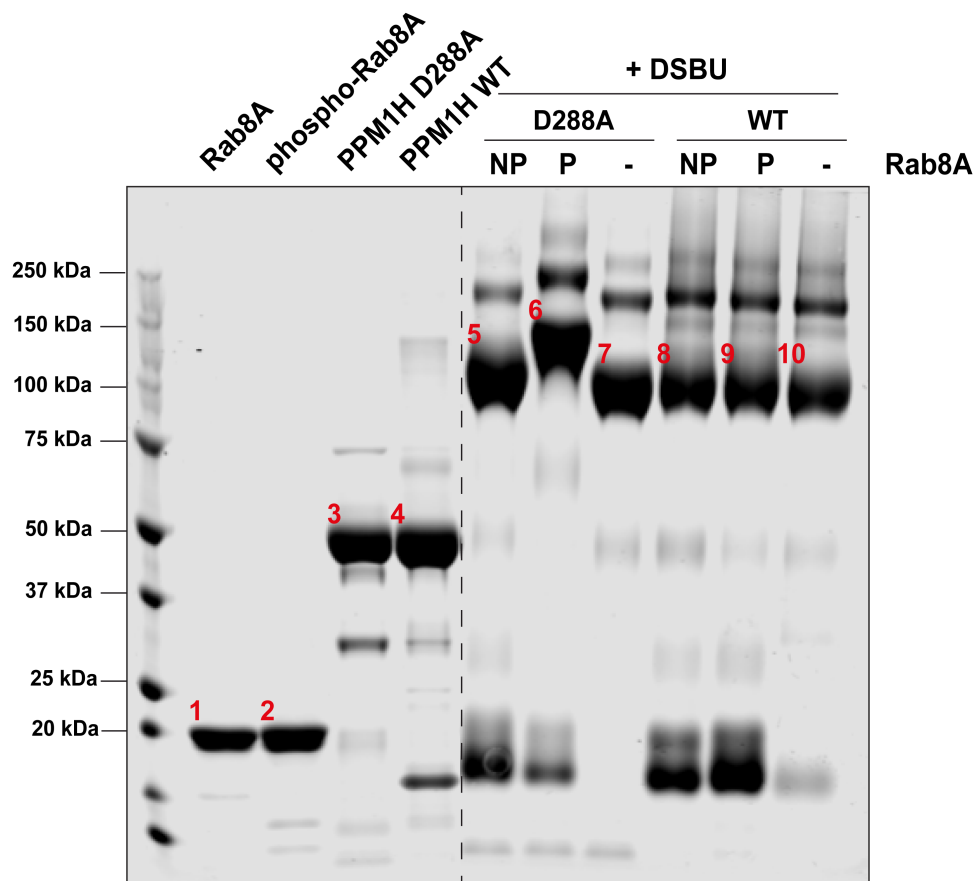


Figure 1. Coomassie-stained SDS-PAGE gel showing the result of cross-linking of PPM1H D288A with phospho-Rab8A. Protein bands on the gel – 1: non-phopsho-Rab8A; 2: phospho-Rab8A; 3: PPM1H(D288A); 4: PPM1H(WT); 5: Cross-linked dimer of PPM1H(D288A); 6: Cross-linked heterotetramer of PPM1H(D288A) and phospho-Rab8A; 7: Cross-linked dimer of PPM1H(D288A); 8-10: Cross-linked dimer of PPM1H(WT)

2.4. MCX SPE cartridge for cross-linked peptide enrichment

MCX cartridge contains mixed-mode sorbent (reversed-phase and cation exchange) which provides higher selectivity and sensitivity for cross-linked peptides enrichment than traditional SCX cartridge (Schmidt and Sinz 2017). Herein, we applied a similar protocol as previous study to enrich one of our tryptic digested samples (Iacobucci et al. 2018). Take tube labelled D from freeze dried sample from trypsin digest section 2.2.X and add 1 mL aqueous solution containing 4% (v/v) H_3PO_4 .

2.4.1. Place the tube in a water bath sonicator at room temperature for 30 minutes.

2.4.2. Wash the MCX cartridge with 2 mL methanol (no syringe or manifold was applied; according to the nature protocol (Iacobucci et al. 2018), slow sample loading can enhance the interaction between the cross-linked peptides and cartridge).

Wash the MCX cartridge with 2 aqueous solution containing 4% (v/v) H_3PO_4

- 2.4.3. Load sample from step 2.3.2 using a 1000 μL pipette onto the washed MCX cartridge.
- 2.4.4. Wash the MCX cartridge with 500 μL aqueous solution containing 4% (v/v) H_3PO_4
- 2.4.5. Wash the MCX cartridge with 500 μL of 10% (v/v) MeOH solution with 0.1% (v/v) FA.
- 2.4.6. Elute low-charged peptides by washing the MCX cartridge with 500 μL solution composed of 500 mM NH_4OAc in 40% (v/v) MeOH solution with 0.1% (v/v) FA.
- 2.4.7. Elute high-charged peptides by washing the MCX cartridge with 700 μL solution composed of 2000 mM NH_4OAc in 80% (v/v) MeOH solution with 0.1% (v/v) FA.
- 2.4.8. Freeze the samples on dry ice
- 2.4.9. Vacuum dry using SpeedVac

2.5. LC MS/MS experiment

Liquid chromatography tandem mass spectrometry (LC MS/MS) experiment was performed on an Ultimate 3000 RSLC nano-HPLC system coupled to an Orbitrap ExplorisTM 480 mass spectrometer.

- 2.5.1. Re-suspend all vacuum dried samples in 30 μL solution containing 3% (v/v) acetonitrile and 0.1% (v/v) FA
- 2.5.2. Place tube in a water bath sonicator at room temperature for 30 minutes
- 2.5.3. Load 3 – 14 μL solution from each sample onto the nano-HPLC system individually. Peptides were trapped by a precolumn (Acclaim PepMapTM 100, C18, 100 μm x 2 cm, 5 μm , 100 $\text{\AA}$) using aqueous solution containing 0.1% (v/v) TFA. Peptides were then separated by an analytical column (PepMapTM RSLC C18, 75 μm x 50 cm, 2 μm , 100 $\text{\AA}$) at 45°C using a linear gradient of 1 to 35% solvent B (solution containing 80% (v/v) acetonitrile and 0.1% (v/v) FA) for 90 minutes, 35 to 85% (v/v) solvent B for 5 minutes, 85% (v/v) solvent B for 10 minutes, 85 to 1% (v/v) solvent B for 1 minute, and 1% (v/v) solvent B for 14 minutes. The flow rate was set at 300 nL/min for all experiments.
- 2.5.4. Data were acquired with data-dependent MS/MS mode. For each MS scan, the scan range was set between 375 and 1500 m/z with the resolution at 120,000 and 300% automatic gain control (AGC) was used. The maximum injection time for each MS scan was 100 ms. The 10 highest abundant peptides with charge state between 2 and 8 as well as intensity threshold higher than $1.0\text{e}+4$ were then isolated with a 1.2 Da isolation window sequentially. Stepped HCD with normalized collision energy of 27, 30, and 33% was applied to fragment the isolated peptides. For each MS/MS scan, the resolution was set at 15,000 with a normalized AGC at 200%. The maximum injection time was set at 250 ms.

Dynamic exclusion with 60 s duration and 2 ppm window was enabled for the experiment.

2.6. Cross-linked peptide data analysis

- 2.6.1. The .RAW files obtained from the LC MS/MS experiments were converted into .mgf files using RawConverter software (He et al. 2015)
- 2.6.2. The .mgf files were submitted to search using MeroX software against PPM1H and Rab-8A protein sequences to identify potential cross-linked peptides (He et al. 2015). Digestive enzyme, trypsin, trypsin and AspN, or trypsin and GluC, were selected according to the experimental setup. 3 maximum missed cleavages with peptide length ranged from 3 to 50 were applied. Carbamidomethylation at Cysteine residue was set as fixed modification; while oxidation at methionine residue and deamidation at asparagine residue were included in variable modification. DSBU cross-linker was selected with specificity cross-linked sites at Lysine, Serine, Threonine, and Tyrosine residues. 10 ppm and 20 ppm were used to filter the mass error in precursor ion (MS1) and fragment ion (MS2) scans. Only ions with signal-to-noise ratio high than 2 were used for database search. RISEUP searching mode was applied, minimum 2 fragments per peptide and 5% false discovery rate (FDR) were required for a cross-linked peptide identification.
- 2.6.3. Potential cross-linked peptides with score higher than 50 were then manually verified to guarantee the cleavage information obtained from the MS experiment (Figure 2 and 3).

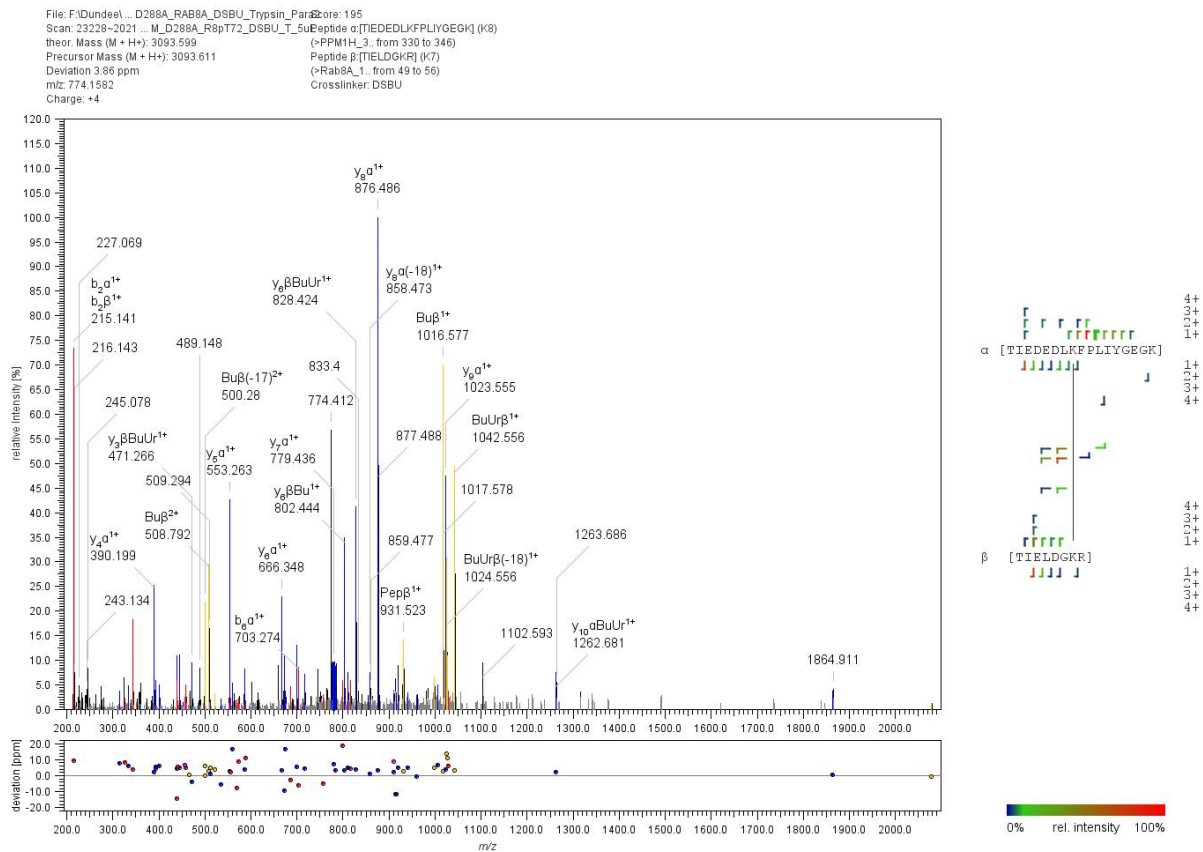


Figure 2. High confident crosslinked PPM1H and pRab8A peptides were identified in the tryptic digested sample with calculated score (by meroX software) equal to 181 and FDR < 0.05. Fragments obtained from the MS spectrum proposed the interaction was located at K8 (PPM1H peptide) and K7 residue (pRab8A peptide). We accepted this crosslinked peptide and included this information in our further analysis since a single interaction site was proposed from each peptide.

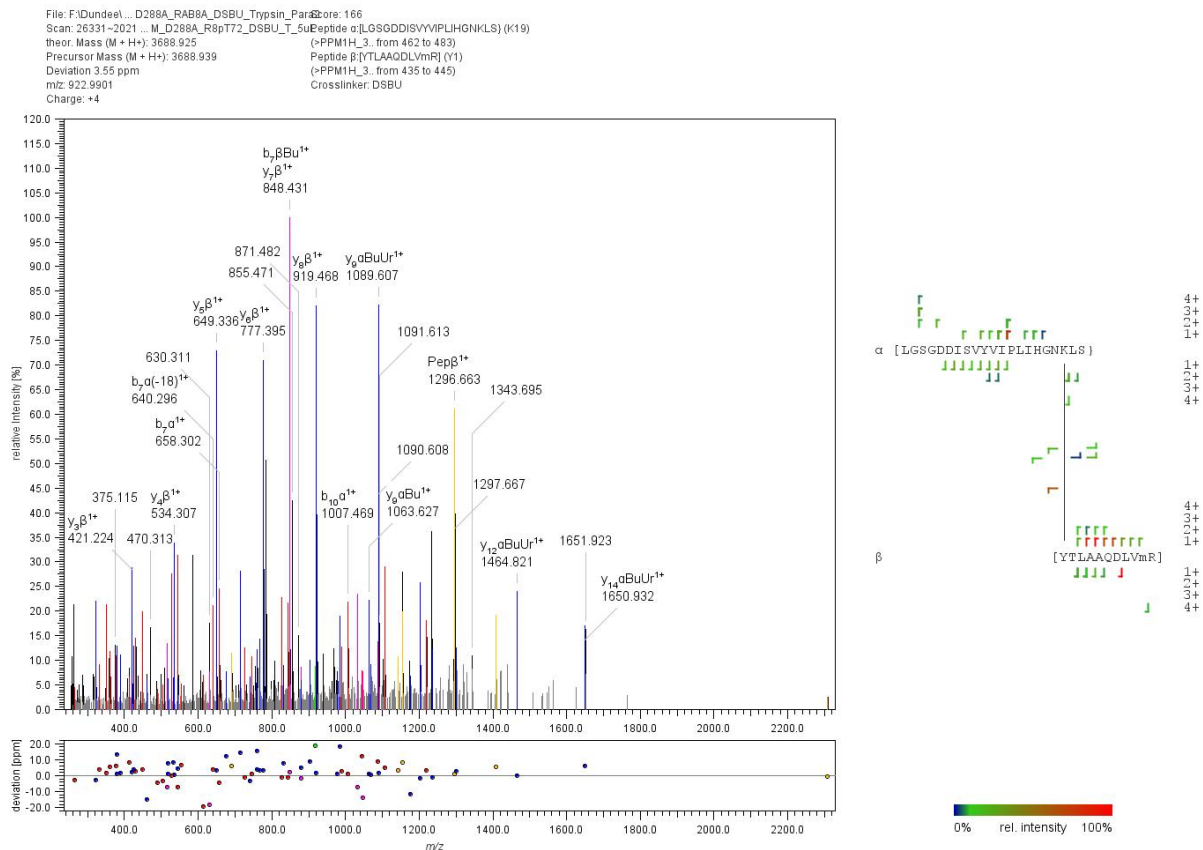


Figure 3. Ambiguous crosslinked PPM1H and pRab8A peptides were found in the tryptic digested sample with the calculated score (by meroX software) equal to 151 and FDR < 0.05. Fragments obtained from the MS spectrum proposed the interaction was located at K19 residue from PPM1H peptide; the fragmentation information for pRab8A peptide, however, could not differentiate the interaction between Y1 and T2 residues. In this situation, we rejected the peptide and excluded this crosslink information for the further analysis.

References

- Berndsen K, Lis P, Yeshaw WM, Wawro PS, Nirujogi RS, Wightman M, Macartney T, Dorward M, Knebel A, Tonelli F, Pfeffer SR, Alessi DR. PPM1H phosphatase counteracts LRRK2 signaling by selectively dephosphorylating Rab proteins. *Elife*. 2019 Oct 30;8:e50416. doi: 10.7554/eLife.50416. PMID: 31663853.
- Götze M, Iacobucci C, Ihling CH, Sinz A. A Simple Cross-Linking/Mass Spectrometry Workflow for Studying System-wide Protein Interactions. *Anal Chem*. 2019 Aug 6;91(15):10236-10244. doi: 10.1021/acs.analchem.9b02372. Epub 2019 Jul 19. PMID: 31283178.
- Götze M, Pettelkau J, Schaks S, Bosse K, Ihling CH, Krauth F, Fritzsche R, Kühn U, Sinz A. StavroX--a software for analyzing crosslinked products in protein interaction studies. *J Am Soc Mass Spectrom*. 2012 Jan;23(1):76-87. doi: 10.1007/s13361-011-0261-2. Epub 2011 Oct 25. PMID: 22038510.

He L, Diedrich J, Chu YY, Yates JR 3rd. Extracting Accurate Precursor Information for Tandem Mass Spectra by RawConverter. *Anal Chem*. 2015 Nov 17;87(22):11361-7. doi: 10.1021/acs.analchem.5b02721. Epub 2015 Nov 4. PMID: 26499134.

Iacobucci C, Götze M, Ihling CH, Piotrowski C, Arlt C, Schäfer M, Hage C, Schmidt R, Sinz A. A cross-linking/mass spectrometry workflow based on MS-cleavable cross-linkers and the MeroX software for studying protein structures and protein-protein interactions. *Nat Protoc*. 2018 Dec;13(12):2864-2889. doi: 10.1038/s41596-018-0068-8. PMID: 30382245.

Pan D, Brockmeyer A, Mueller F, Musacchio A, Bange T. Simplified Protocol for Cross-linking Mass Spectrometry Using the MS-Cleavable Cross-linker DSBU with Efficient Cross-link Identification. *Anal Chem*. 2018 Sep 18;90(18):10990-10999. doi: 10.1021/acs.analchem.8b02593. Epub 2018 Aug 29. PMID: 30074391.

Schmidt R, Sinz A. Improved single-step enrichment methods of cross-linked products for protein structure analysis and protein interaction mapping. *Anal Bioanal Chem*. 2017 Mar;409(9):2393-2400. doi: 10.1007/s00216-017-0185-1. Epub 2017 Jan 12. PMID: 28083664.

Steger M, Diez F, Dhekne HS, Lis P, Nirujogi RS, Karayel O, Tonelli F, Martinez TN, Lorentzen E, Pfeffer SR, Alessi DR, Mann M. Systematic proteomic analysis of LRRK2-mediated Rab GTPase phosphorylation establishes a connection to ciliogenesis. *Elife*. 2017 Nov 10;6:e31012. doi: 10.7554/eLife.31012. PMID: 29125462.